Model Building- GAM Models

Load neccessary packages

```
library(tidyverse)
library(tidymodels)
library(ggplot2)
library(purrr)
library(caret)
library(glmnet)
library(mgcv)
library(splines)
library(pdp)
library(earth)
```

Load in dataset and use 0/1 for binary variables

```
data <- read.csv("data/data_clean.csv") |>
  data.frame()

data$UnablePregnant <- ifelse(data$UnablePregnant == "Yes", 1,
                              ifelse(data$UnablePregnant == "No", 0, NA))

data$RegularPeriod_12m <- ifelse(data$RegularPeriod_12m == "Yes", 1,
                              ifelse(data$RegularPeriod_12m == "No", 0, NA))

data = data |>
  mutate(
    UnablePregnant = as.factor(UnablePregnant),
    RegularPeriod_12m = as.factor(RegularPeriod_12m)
  )
```

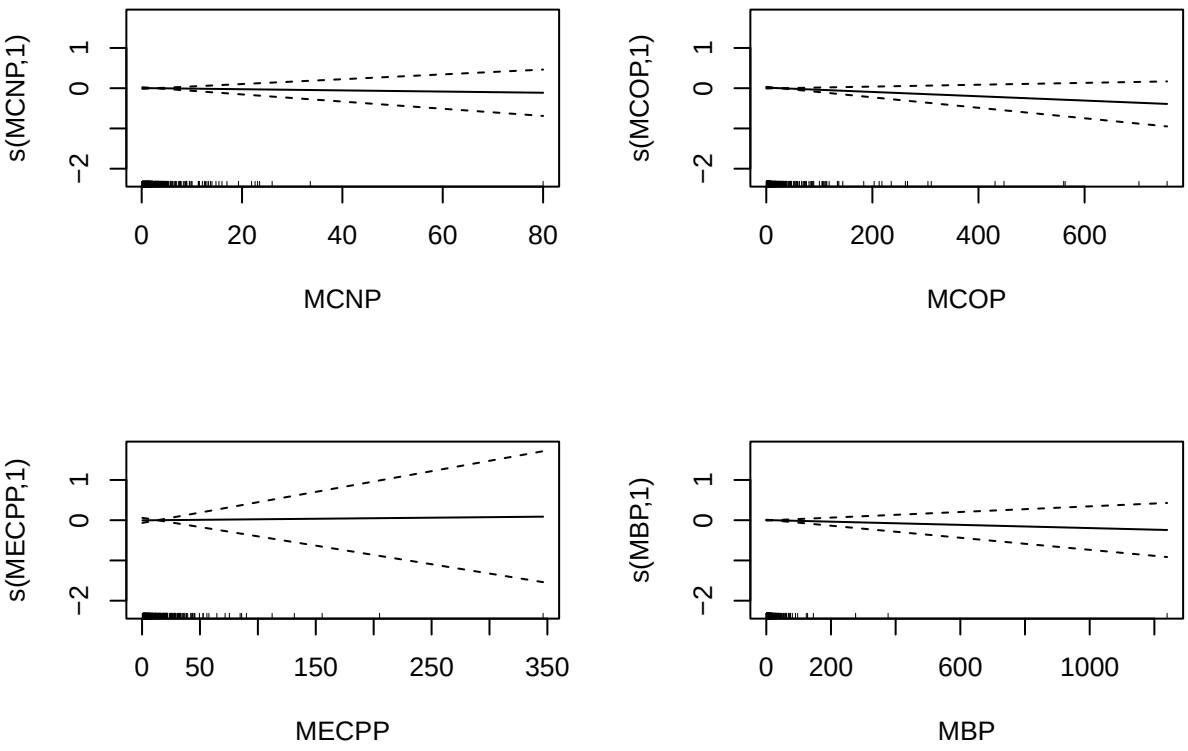
Model using outome as Age_FirstPeriod (counts)

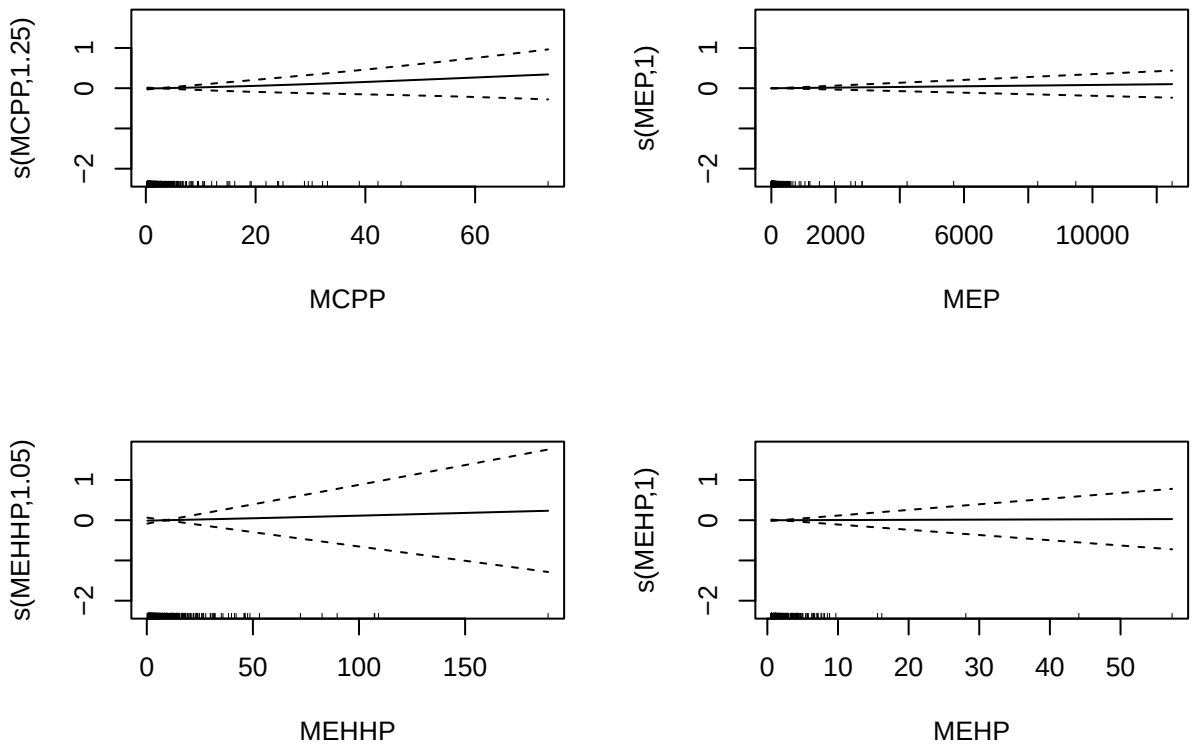
```
# Model Fitting using mgcv

# Fit model
gam.m1 <- gam(Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) + s(MBP) +
                s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
                Year,
```

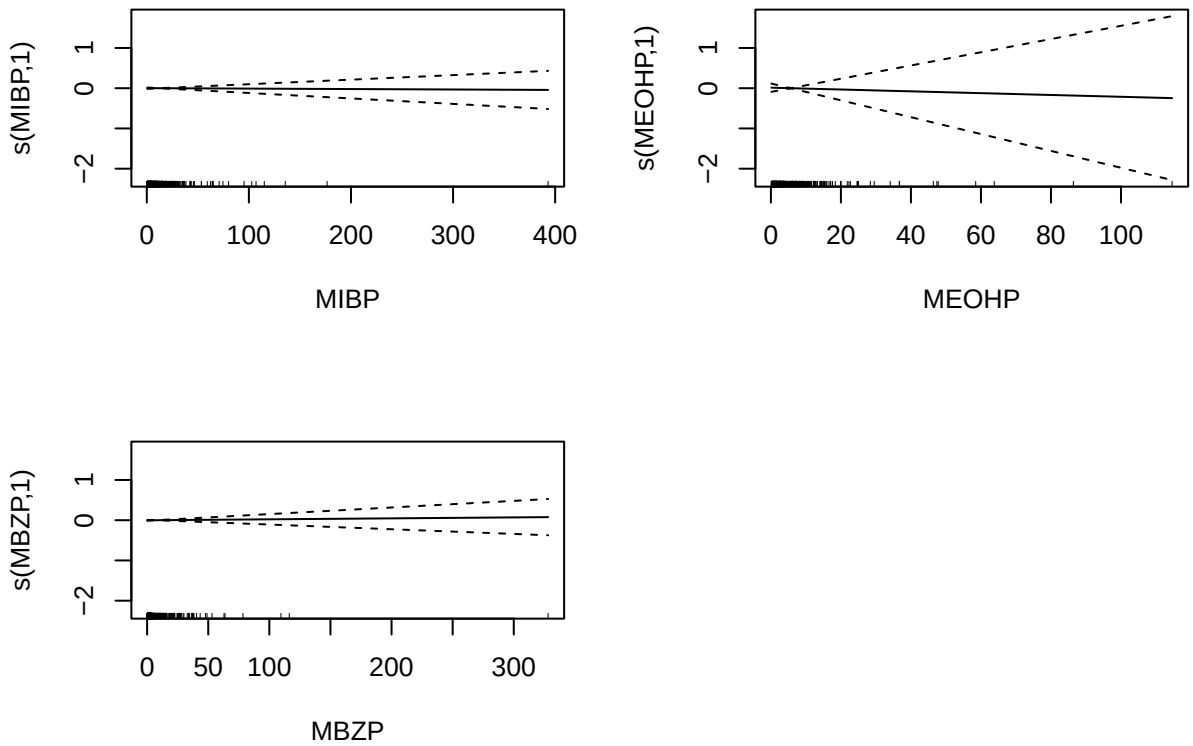
```r
              data = data,
              family = poisson)
summary(gam.m1)
```

```
##
## Family: poisson
## Link function: log
##
## Formula:
## Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) +
##     s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##     s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)                2.5038439  0.0826413  30.298   <2e-16 ***
## Age                        0.0009202  0.0014594   0.631    0.528
## RaceNonHispanic_Black     -0.0051571  0.0416891  -0.124    0.902
## RaceNonHispanic_White     -0.0329513  0.0398012  -0.828    0.408
## RaceOther_Hispanic        -0.0458747  0.0497334  -0.922    0.356
## RaceOtherRace_MultiRacial  0.0106291  0.0441375   0.241    0.810
## Year2015_2016             -0.0081618  0.0311574  -0.262    0.793
## Year2017_2018              0.0100002  0.0318397   0.314    0.753
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##            edf Ref.df Chi.sq p-value
## s(MCNP)  1.000  1.000  0.147   0.702
## s(MCOP)  1.000  1.000  1.872   0.171
## s(MECPP) 1.000  1.000  0.011   0.916
## s(MBP)   1.000  1.000  0.503   0.478
## s(MCPP)  1.247  1.455  1.465   0.454
## s(MEP)   1.000  1.000  0.346   0.557
## s(MEHHP) 1.054  1.106  0.118   0.852
## s(MEHP)  1.000  1.000  0.005   0.942
## s(MIBP)  1.000  1.000  0.031   0.861
## s(MEOHP) 1.000  1.000  0.056   0.813
## s(MBZP)  1.000  1.000  0.111   0.739
##
## R-sq.(adj) =  0.0306   Deviance explained = 6.55%
## UBRE = -0.66611  Scale est. = 1         n = 513
```

```r
# Generate plots for the nonlinear terms included in the model
plot(gam.m1, pages=4)
```


s(MCNP,1)
1
0
−2
0
20
40
60
80
MCNP
s(MCOP,1)
1
0
−2
0
200
400
600
MCOP
s(MECPP,1)
1
0
−2
0
50
150
250
350
MECPP
s(MBP,1)
1
0
−2
0
200
600
1000
MBP


s(MCPP,1.25)
1
0
−2
0
20
40
60
MCPP
s(MEP,1)
1
0
−2
0
2000
6000
10000
MEP
s(MEHHP,1.05)
1
0
−2
0
50
100
150
MEHHP
s(MEHP,1)
1
0
−2
0
10
20
30
40
50
MEHP



Model using outome as Age_FirstPeriod (counts)

```
# Model Training using caret

# Matrix of predictors
x1 <- model.matrix(Age_FirstPeriod ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                     MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]
# Vector of response
y1 <- data$Age_FirstPeriod
# Set up CV
ctrl1 <- trainControl(method = "cv", number = 10)
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit1 <- train(x1, y1,
                  method = "gam",
                  family = poisson,
                  trControl = ctrl1)
# Best tuning parameters
gam.fit1$bestTune
# Final model
gam.fit1$finalModel
# Generate plots for the nonlinear terms included in the model
```

```r
plot(gam.fit1$finalModel, pages=4)
```

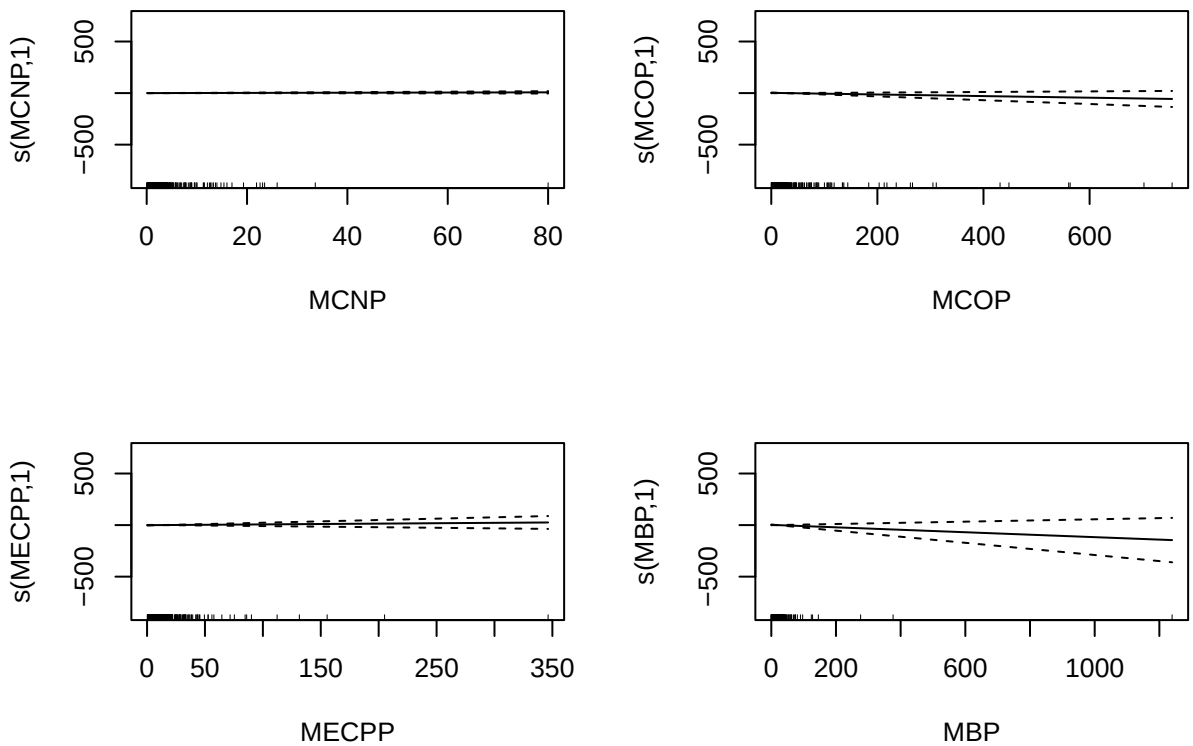
Model using outome as RegularPeriod_12m (binary)

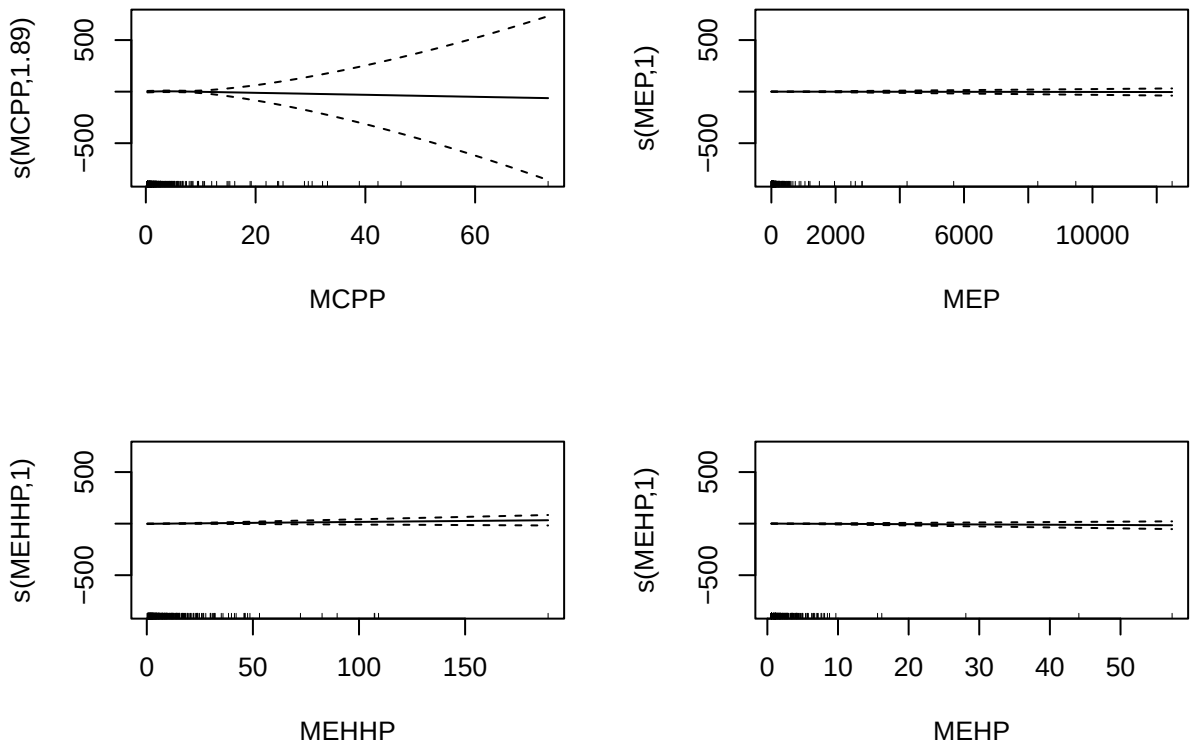
```r
# Model Fitting using mgcv

# Fit model
gam.m2 <- gam(RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) + s(MBP) +
                s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
                Year,
              data = data,
              family = binomial)
summary(gam.m2)
```

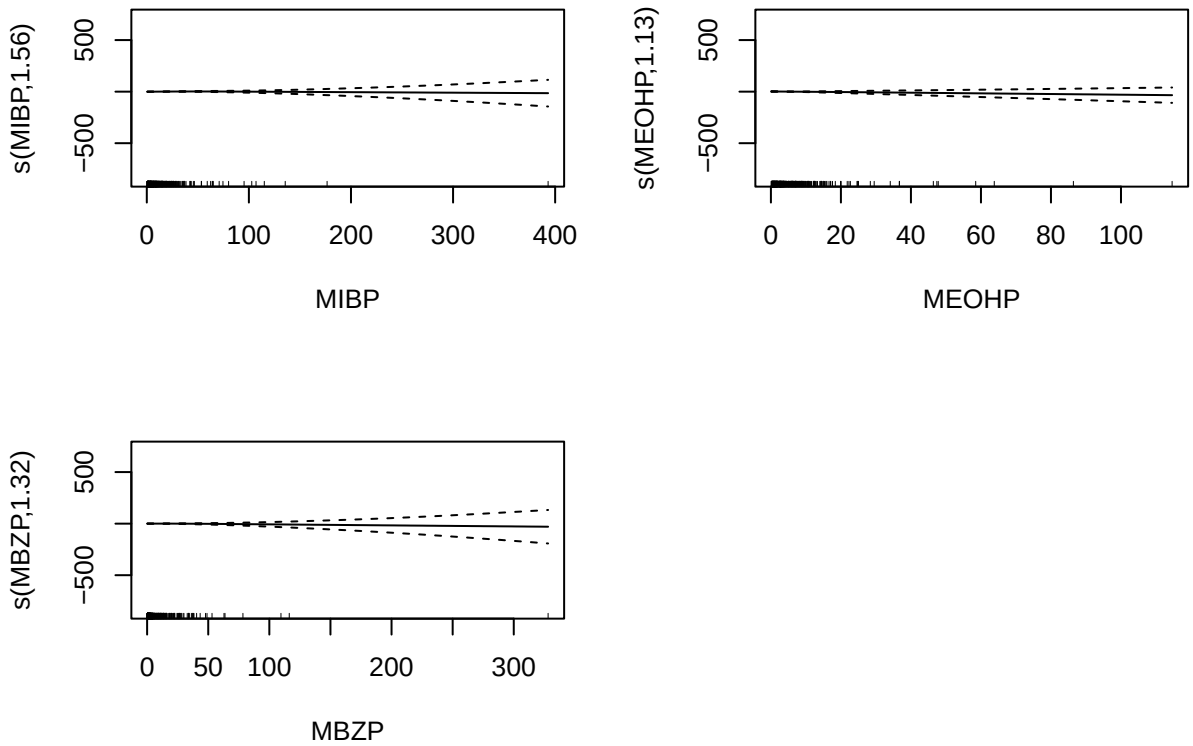
```
##
## Family: binomial
## Link function: logit
##
## Formula:
## RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) +
##     s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##     s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##                             Estimate Std. Error z value Pr(>|z|)
## (Intercept)               -6.914e+01  7.599e+06   0.000 0.999993
## Age                       -1.348e-01  3.714e-02  -3.631 0.000283 ***
## RaceNonHispanic_Black      6.845e+01  7.599e+06   0.000 0.999993
## RaceNonHispanic_White      6.781e+01  7.599e+06   0.000 0.999993
## RaceOther_Hispanic         6.911e+01  7.599e+06   0.000 0.999993
## RaceOtherRace_MultiRacial  6.871e+01  7.599e+06   0.000 0.999993
## Year2015_2016             -9.912e-01  1.186e+00  -0.836 0.403245
## Year2017_2018             -2.967e-01  1.074e+00  -0.276 0.782302
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##            edf Ref.df Chi.sq p-value
## s(MCNP)  1.000  1.000  1.564   0.211
## s(MCOP)  1.000  1.000  2.056   0.152
## s(MECPP) 1.000  1.000  0.636   0.425
## s(MBP)   1.000  1.000  1.758   0.185
## s(MCPP)  1.891  2.218  3.128   0.172
## s(MEP)   1.000  1.000  0.046   0.830
## s(MEHHP) 1.000  1.000  1.633   0.201
## s(MEHP)  1.000  1.000  0.648   0.421
## s(MIBP)  1.560  1.909  1.295   0.547
## s(MEOHP) 1.130  1.232  0.611   0.448
## s(MBZP)  1.324  1.560  0.244   0.740
##
```

```
## R-sq.(adj) =   0.27   Deviance explained = 40.3%
## UBRE = -0.81303  Scale est. = 1         n = 513
```

```r
# Generate plots for the nonlinear terms included in the model
plot(gam.m2, pages=4)
```




s(MCPP,1.89)
500
−500
0
20
40
60
MCPP
s(MEP,1)
500
−500
0
2000
6000
10000
MEP
s(MEHHP,1)
500
−500
0
50
100
150
MEHHP
s(MEHP,1)
500
−500
0
10
20
30
40
50
MEHP



Model using outome as RegularPeriod_12m (binary)

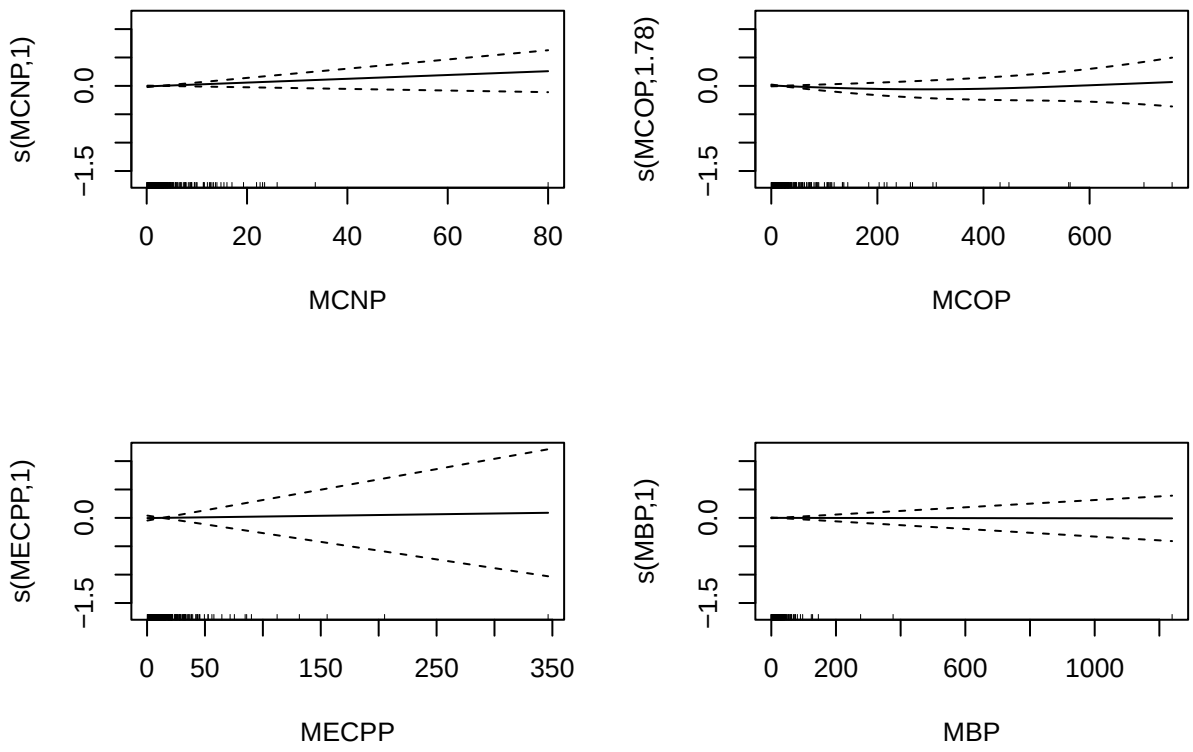
```
# Model Training using caret

# Matrix of predictors
x2 <- model.matrix(RegularPeriod_12m ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                     MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]
# Vector of response
y2 <- data$RegularPeriod_12m
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit2 <- train(x2, y2,
                  method = "gam",
                  family = binomial,
                  trControl = ctrl1)
# Best tuning parameters
gam.fit2$bestTune
# Final model
gam.fit2$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit2$finalModel, pages=4)
```

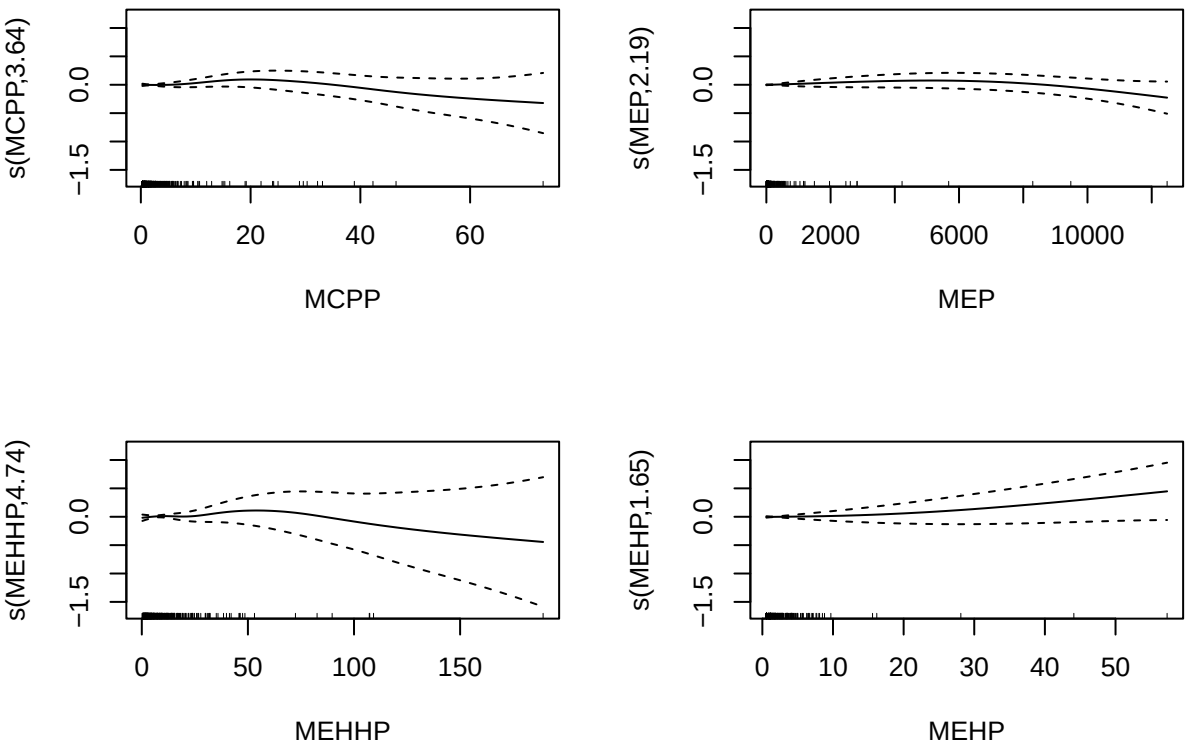
Model using outome as Age_LastPeriod (counts)

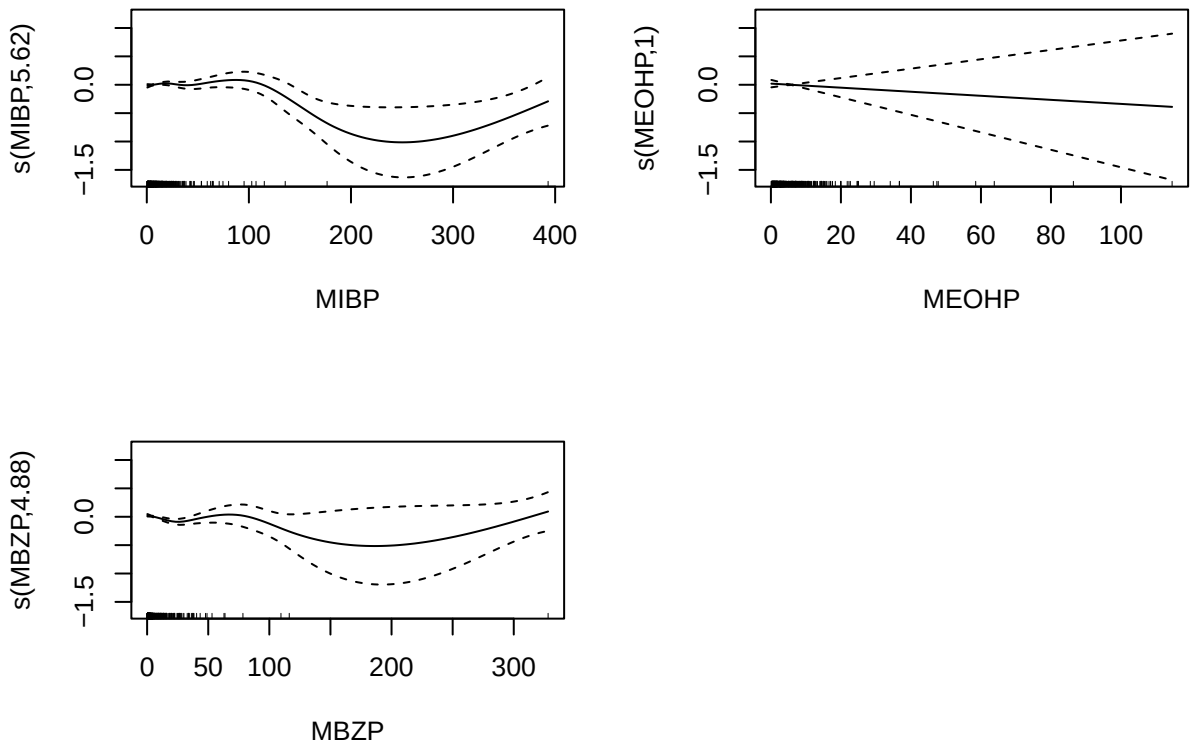
```r
# Model Fitting using mgcv

# Fit model
gam.m3 <- gam(Age_LastPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) + s(MBP) +
                s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
                Year,
              data = data,
              family = poisson)
summary(gam.m3)
```

```
##
## Family: poisson
## Link function: log
##
## Formula:
## Age_LastPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) +
##     s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##     s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##                              Estimate Std. Error z value Pr(>|z|)
## (Intercept)                 2.7363206  0.0528609  51.765   <2e-16 ***
## Age                         0.0194870  0.0009437  20.650   <2e-16 ***
## RaceNonHispanic_Black      -0.0170992  0.0237875  -0.719   0.4722
## RaceNonHispanic_White      -0.0203023  0.0230419  -0.881   0.3783
## RaceOther_Hispanic         -0.0049640  0.0274016  -0.181   0.8562
## RaceOtherRace_MultiRacial   0.0497177  0.0246003   2.021   0.0433 *
## Year2015_2016               0.0053107  0.0179284   0.296   0.7671
## Year2017_2018               0.0106134  0.0182004   0.583   0.5598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##            edf Ref.df Chi.sq p-value
## s(MCNP)  1.000  1.000  1.883 0.17003
## s(MCOP)  1.777  2.161  1.754 0.47153
## s(MECPP) 1.000  1.000  0.025 0.87520
## s(MBP)   1.000  1.000  0.002 0.96443
## s(MCPP)  3.643  4.404  6.569 0.21996
## s(MEP)   2.189  2.722  2.495 0.29941
## s(MEHHP) 4.743  5.669  7.458 0.30245
## s(MEHP)  1.650  1.985  3.562 0.19189
## s(MIBP)  5.622  6.588 18.718 0.00701 **
## s(MEOHP) 1.000  1.000  0.350 0.55391
## s(MBZP)  4.877  5.742 16.337 0.00799 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.505   Deviance explained = 55.9%
## UBRE = 0.11635  Scale est. = 1         n = 513
```

```r
# Generate plots for the nonlinear terms included in the model
plot(gam.m3, pages=4)
```




s(MCPP,3.64)
0.0
−1.5
0
20
40
60
MCPP
s(MEP,2.19)
0.0
−1.5
0
2000
6000
10000
MEP
s(MEHHP,4.74)
0.0
−1.5
0
50
100
150
MEHHP
s(MEHP,1.65)
0.0
−1.5
0
10
20
30
40
50
MEHP



Model using outome as Age_LastPeriod (counts)

```
# Model Training using caret

# Matrix of predictors
x3 <- model.matrix(Age_LastPeriod ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                     MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]
# Vector of response
y3 <- data$Age_LastPeriod
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit3 <- train(x3, y3,
                  method = "gam",
                  family = poisson,
                  trControl = ctrl1)
# Best tuning parameters
gam.fit3$bestTune
# Final model
gam.fit3$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit3$finalModel, pages=4)
```

Model using outome as UnablePregnant (binary)

```r
# Model Fitting using mgcv

# Fit model
gam.m4 <- gam(UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) + s(MBP) +
                s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
                Year,
              data = data,
              family = binomial)
```

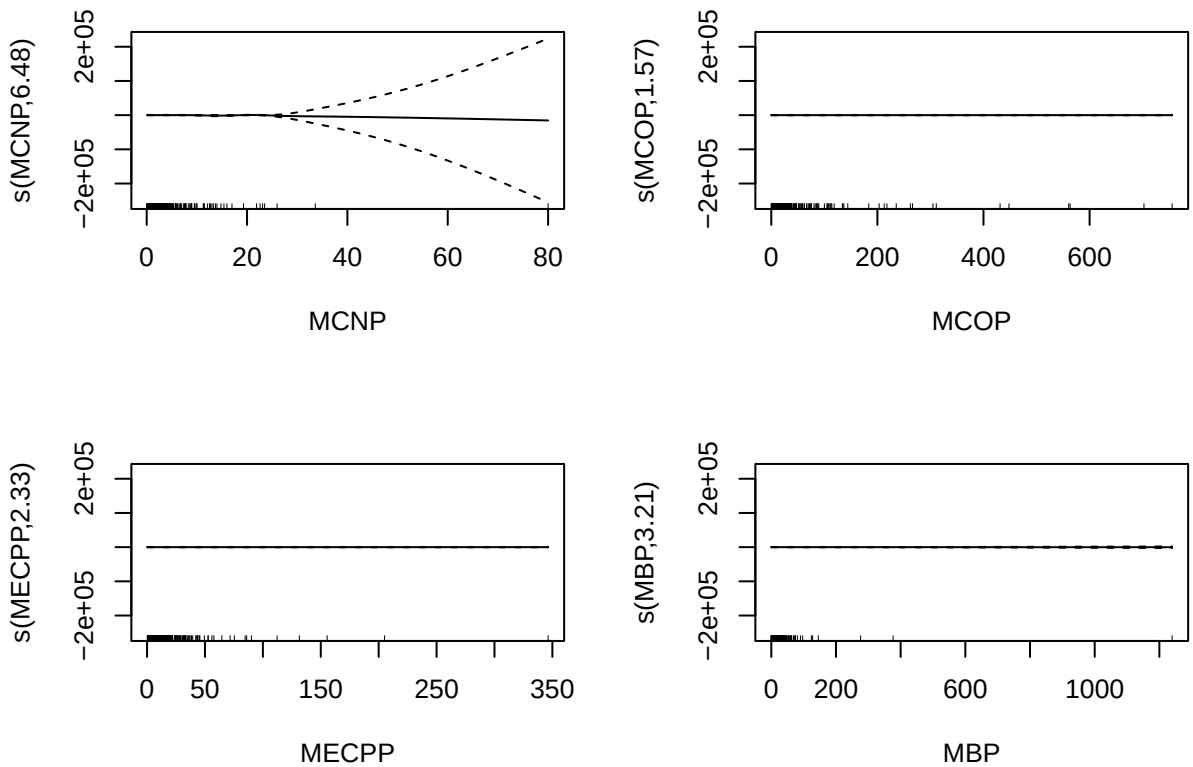
```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Iteration limit reached without full convergence - check carefully
```

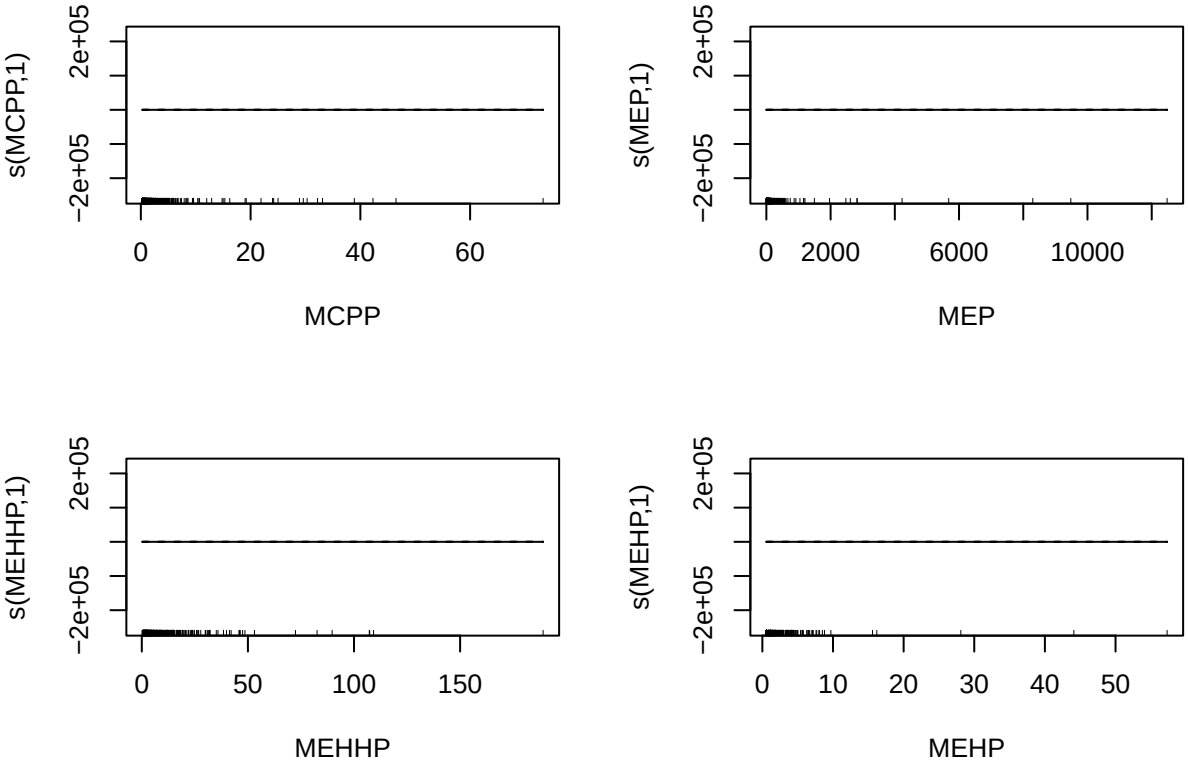
```r
summary(gam.m4)
```

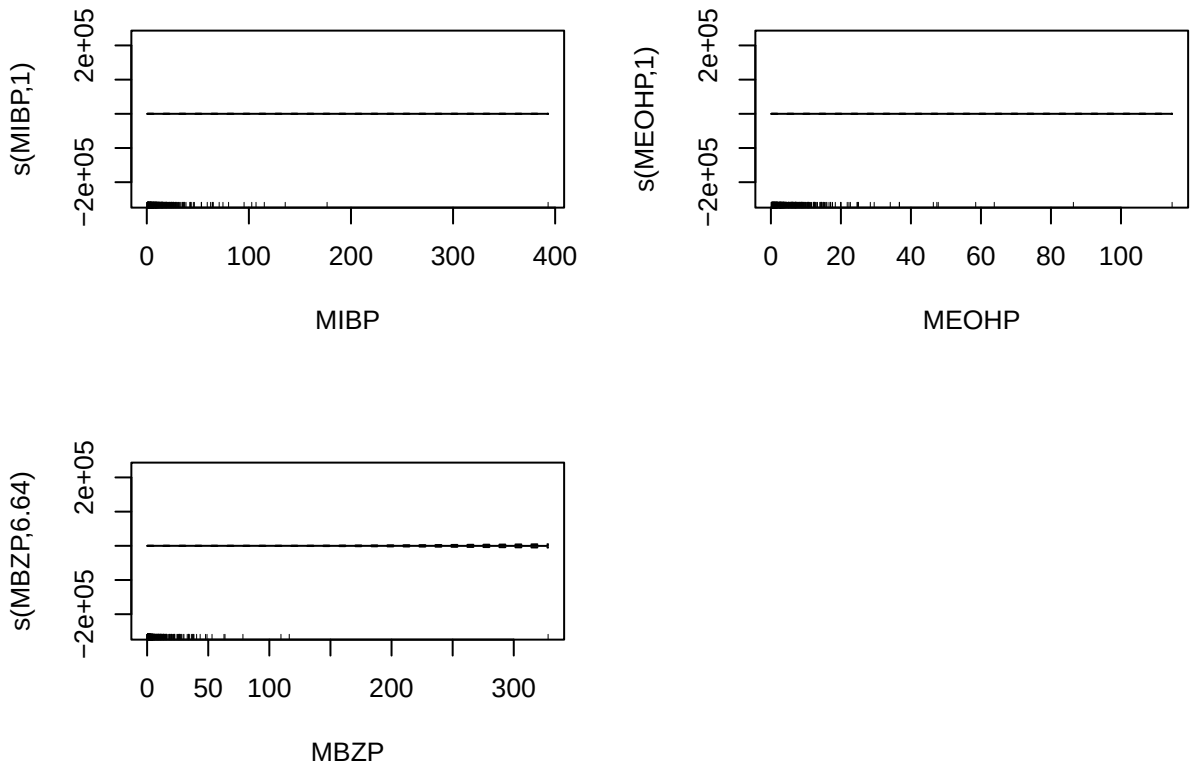
```
##
## Family: binomial
## Link function: logit
##
## Formula:
## UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECPP) +
##     s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##     s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)               -87.61561  258.64433  -0.339   0.7348
## Age                         0.04320    0.02501   1.727   0.0842 .
## RaceNonHispanic_Black       1.58233    0.65824   2.404   0.0162 *
## RaceNonHispanic_White       1.19291    0.66064   1.806   0.0710 .
## RaceOther_Hispanic          0.37795    0.79320   0.476   0.6337
## RaceOtherRace_MultiRacial   0.98160    0.67896   1.446   0.1483
## Year2015_2016               0.53329    0.42480   1.255   0.2093
## Year2017_2018               0.34498    0.46300   0.745   0.4562
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##            edf Ref.df Chi.sq p-value
## s(MCNP)  6.482  6.737  3.133   0.858
## s(MCOP)  1.568  1.932  1.097   0.581
## s(MECPP) 2.332  2.832  4.114   0.143
## s(MBP)   3.214  3.787  4.170   0.251
## s(MCPP)  1.000  1.000  1.778   0.182
## s(MEP)   1.000  1.000  0.442   0.506
## s(MEHHP) 1.000  1.000  0.311   0.577
## s(MEHP)  1.000  1.000  0.231   0.631
## s(MIBP)  1.000  1.000  0.111   0.739
## s(MEOHP) 1.000  1.000  0.054   0.816
## s(MBZP)  6.636  6.934  9.739   0.196
```

```
##
## R-sq.(adj) =  0.122   Deviance explained = 23.2%
## UBRE = -0.36908  Scale est. = 1         n = 513
```

```r
# Generate plots for the nonlinear terms included in the model
plot(gam.m4, pages=4)
```




s(MCPP,1)
2e+05
−2e+05
0
20
40
60
MCPP
s(MEP,1)
2e+05
−2e+05
0
2000
6000
10000
MEP
s(MEHHP,1)
2e+05
−2e+05
0
50
100
150
MEHHP
s(MEHP,1)
2e+05
−2e+05
0
10
20
30
40
50
MEHP



Model using outome as UnablePregnant (binary)

```
# Model Training using caret

# Matrix of predictors
x4 <- model.matrix(UnablePregnant ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                     MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]
# Vector of response
y4 <- data$UnablePregnant
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit4 <- train(x4, y4,
                  method = "gam",
                  family = binomial,
                  trControl = ctrl1)
# Best tuning parameters
gam.fit4$bestTune
# Final model
gam.fit4$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit4$finalModel, pages=4)
```